



Case Study II: Next-Lactation Milk Yield Forecasting

Nixon M Paulson
Rohith B Chowdappa
Maria Cherian

Problem Statement

- ▶ Predict **total milk yield in the next lactation**
- ▶ Setup: features from lactation $n \rightarrow$ target = `mkg_sum( $n+1$ )`
- ▶ Data: OptiKuh daily dairy records (1,714 cows, $\sim 3,400$ lactations)
- ▶ Aggregation level: one row per cow (1om) + lactation (1nr)

Initial Challenges

- ▶ Daily records must be aggregated without mixing lactations
- ▶ Disease fields are categorical text, not numeric flags
- ▶ Most cows appear in only one or two lactation episodes

Feature Engineering Approach

- ▶ Sort records by cow, lactation, and observation date
- ▶ Aggregate to lactation level with mean / min / max:
 - ▶ Nutrition: `fts`, `ftsnel`, `ftsme`, `eb_nel`, `eb_me`
 - ▶ Physiology: `bcs`, `gew`, blood markers (`bhb`, `nefa`, `glukose`)
- ▶ Milk signals: `mkg_sum`, `mkg_mean`, `mkg_max`
- ▶ Target: `groupby(lom) ["mkg_sum"].shift(-1)`; drop last lactation per cow
- ▶ All model inputs forced to numeric (`float64`) before training

Disease Handling

Initial approach

- ▶ One-hot encoding of diagnosis text (26 sparse boolean columns)
- ▶ High dimensionality, low signal

Improved approach

- ▶ `disease_count`: total disease events per lactation
- ▶ `had_disease`: binary flag (any event)
- ▶ `disease_category_count`: number of disease groups affected

Models Used

- ▶ **HistGradientBoostingRegressor**
- ▶ Handles missing values natively
- ▶ Hyperparameters: `max_depth=8`, `learning_rate=0.05`,
`max_iter=500`
- ▶ Baseline and improved pipelines use the same model for fair comparison

Key Issue Discovered

- ▶ **Missing temporal structure** was the main failure mode
- ▶ 74% of cows contributed only one supervised row → lag features mostly empty
- ▶ 37% of lactations had `mkg_sum` = 0 (incomplete milk recording window)
- ▶ Time-based split left early train folds without usable history
- ▶ Baseline excluded current-lactation `mkg_sum` from features

To note: next-lactation yield needs cow-level milk history, not only within-lactation averages.

Improvements Added

- ▶ Lag features per cow (ordered by calving date):
 - ▶ `prev_lactation_milk`, `prev_prev_lactation_milk`
 - ▶ `milk_trend`, `cow_avg_milk` (prior lactations only)
- ▶ Drop lactations with zero recorded milk (current or next)
- ▶ Simplified disease counts instead of one-hot explosion
- ▶ Remove all-NaN / constant columns before fitting

Results

	RMSE (kg)	MAE (kg)	R²
Baseline (before improvements)	3,291	2,494	0.066
Improved model	2,556	1,941	0.453

- ▶ 44 numeric features in final model
- ▶ Largest gain from lag features + data quality, not model complexity

Conclusions & Next Steps

Conclusions

- ▶ Correct temporal structure matters more than adding many raw features
- ▶ Simple sklearn baseline can reach useful performance with clean aggregation

Next steps

- ▶ Tune features for cows with longer lactation histories
- ▶ Use designated time series models for better temporal modeling?